Does Your AI Benchmark Measure Intelligence—
or Just Memory?

Building an interactive lab to stress-test whether ARC-AGI performance survives freshly generated tasks

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The Uncomfortable Question

Imagine you built a model that scores 85% on a well-known AI reasoning benchmark. You announce the result, the community celebrates, and the leaderboard updates. But here is the uncomfortable truth: **that benchmark has been publicly available since 2019**. Every task, every input-output pair, every transformation rule—indexed, crawled, and ingested into the very training corpora your model learned from.

So when your model "solves" a benchmark task, is it genuinely reasoning about abstract patterns? Or is it retrieving a familiar answer from its training data? This is the question that keeps benchmark researchers up at night—and it is the question that the **ARC Generalization Lab** was built to investigate.

The Contamination Problem

The Abstraction and Reasoning Corpus (ARC-AGI), created by François Chollet in 2019, is widely regarded as one of the most challenging AI benchmarks. Each task presents a few input → output grid pairs that demonstrate a transformation rule. The test: infer the rule and apply it to a new input. No natural language description is provided. Pure visual pattern recognition and rule induction.

The problem? **Bordes et al. (2024)** showed empirically that ARC-AGI evaluation tasks appear verbatim in Common Crawl and The Pile—the massive web-scraped datasets used to train modern LLMs. This means any model trained on these corpora has likely "seen" the test set. Its score may reflect familiarity rather than genuine generalization.

> *"An In-Depth Look at Gemini's ARC-AGI Capabilities and Contamination revealed that significant portions of ARC evaluation data appear in standard pretraining corpora, calling into question whether measured performance reflects true abstract reasoning."*

Our Approach: The Fresh-Task Diagnostic

The ARC Generalization Lab takes a direct approach to this problem. Instead of debating whether contamination matters in theory, we **measure its potential effect empirically**.

The method is straightforward:

- **Step 1:** Take a solver—any solver—and evaluate it on a sample of public ARC-AGI-1 tasks.
- **Step 2:** Generate fresh, never-before-seen ARC-style tasks whose measurable properties (grid size, colour count, pair count) match the public evaluation distribution.
- **Step 3:** Evaluate the exact same solver on these fresh tasks.
- **Step 4:** Compare. If performance drops on fresh tasks, something interesting is happening.

We call this the **public-vs-fresh diagnostic**. The performance gap is not proof of memorization—it is a diagnostic signal consistent with multiple hypotheses: benchmark familiarity, distribution mismatch in unmeasured dimensions, generator artifacts, task ambiguity, or solver baseline limitations. Scientific honesty demands we present all five.

How the Lab Works

The ARC Generalization Lab is a full-stack interactive research instrument:

Frontend (Vite + Vanilla JS)

An interactive single-page application featuring ARC grid rendering, a 5-step guided learner journey, an experiment control panel, a recurrent state sandbox, and a check-your-understanding quiz. Every piece of data displayed is explicitly labelled with its source category.

Backend (FastAPI)

A Python API server that serves public ARC tasks, executes solvers (enhanced heuristic families covering geometric, color remapping, tiling, and border extraction, alongside a random baseline), runs evaluations with 95% Wilson score confidence intervals, and wraps the upstream arc-task-gen pipeline for live generation. A strict evidence-labelling middleware ensures no result is returned without provenance metadata.

Task Generation (pathwaycom/arc-task-gen)

Fresh tasks are generated through a rigorous multi-stage pipeline: distribution analysis of the public evaluation set, joint constraint sampling (preserving natural covariance between grid properties), LLM-based rule generation, structural validation, semantic deduplication, and public-eval similarity filtering. The result: tasks that are measurably similar to public ARC tasks but genuinely novel.

The 60-Second Guided Experience

First-time visitors are guided through a 5-step journey designed to build understanding incrementally:

- **TRY:** Interact with a real ARC task—inspect input/output pairs, run a heuristic solver, reveal ground truth.
- **QUESTION:** Confront the core question: does high public accuracy prove generalization? (Answer: no.)
- **TEST:** Trigger the public-vs-fresh diagnostic comparison using the exact same solver on both task sets.
- **INTERPRET:** Learn that a performance gap is diagnostic evidence, not proof of memorization. Five competing explanations are explicitly presented.
- **ADAPT:** Discover BDH-CQ's approach to test-time adaptation via recurrent associative memory without parameter updates.

BDH-CQ: A Different Kind of Test-Time Adaptation

The benchmark integrity question naturally leads to a deeper one: **how should a model adapt to new tasks at evaluation time?**

Optimization-based approaches like Test-Time Training (Akyürek et al., 2024) perform gradient-based weight updates on each test instance. This is effective—TTT with Llama-3 8B achieves ~53% on ARC—but requires backpropagation infrastructure at inference time and risks catastrophic forgetting.

BDH-CQ (Engdahl et al., 2026) takes the opposite path. Its 150M parameters remain **completely frozen** during evaluation. Instead of updating weights, BDH-CQ accumulates each demonstration into an evolving recurrent associative memory state through forward-pass binding operations. No gradients, no backpropagation, no parameter updates at test time.

The result: 29.5% pass@2 on public ARC-AGI-1 at **$0.0007 per task**—orders of magnitude cheaper than frontier approaches. Our lab includes an interactive toy sandbox where learners can manipulate this recurrent state directly: changing demonstration patterns, adjusting retention rates, and observing how the 4×4 associative state matrix evolves in real time.

Dimension	OpenAI o3	TTT (Akyürek 2024)	BDH-CQ (Engdahl 2026)
ARC-AGI accuracy	75.7% pass@3	~53% (8B)	29.5% pass@2
Test-time weight updates	Undisclosed	Yes (gradients)	None (W frozen)
Cost per task	~$17-33 est.	Moderate	$0.0007
Parameters	Frontier-scale	8B	150M

Table: Architectural comparison across test-time adaptation approaches on ARC-AGI-1.

Radical Transparency: Evidence Labelling

One of our strongest design commitments is that **no number appears without a source label**. Every datum in the interface is explicitly categorized:

- **PUBLISHED RESULT — PATHWAY:** Figures cited directly from Engdahl et al. (2026) or Pathway technical documentation. We did not retrain BDH-CQ or independently verify these numbers.
- **MEASURED:** Computed live by our local solver on real tasks during the experiment.
- **DEMO DATA:** Hand-crafted tutorial tasks (DEMO-001 through DEMO-003), clearly labelled and never presented as experimental evidence.
- **TOY / DIDACTIC:** The in-browser 4×4 recurrent state sandbox, explicitly badged as an educational simplification inspired by BDH-CQ—not the official implementation.

This taxonomy prevents the most common scientific communication failure: presenting derivative or simulated results as if they were primary experimental evidence.

What We Don't Know (And Why That Matters)

Scientific honesty requires us to be explicit about the boundaries of our findings:

- **Solvability is not guaranteed.** Generated tasks pass structural validation but may have ambiguous or unsolvable rules. We mitigate this with blind human audits, but cannot eliminate it.
- **Distribution matching is partial.** We match measurable properties (grid size, colour count, pair count), but cognitive complexity, visual salience, and transformation-type diversity may differ between public and generated tasks in ways we cannot currently measure.
- **Small samples yield wide confidence intervals.** Demo-mode experiments with N=5 or N=10 are exploratory, not definitive. We compute 95% Wilson score confidence intervals on all diagnostic gaps to communicate uncertainty honestly.
- **The gap has multiple possible causes.** We never claim that a performance drop "proves" memorization. Five competing hypotheses are presented with equal weight.
- **BDH-CQ maturity is early-stage.** Developer-reported results exist, but no independent reproduction has been published as of September 2026. All cited figures are clearly labelled as such.

What We Learned Building This

Building the ARC Generalization Lab reinforced three convictions:

First, benchmark integrity matters more than benchmark scores. A leaderboard number is meaningless if we cannot distinguish skill from familiarity. The community needs more diagnostic tools—not just better models.

Second, scientific honesty is a design decision, not an afterthought. Every label, every disclaimer, every alternative hypothesis we surface in the interface was a deliberate choice. It would have been easier to show a single dramatic accuracy gap and call it proof. We chose not to.

Third, interactivity changes understanding. Reading about recurrent associative memory is one thing. Watching a state matrix evolve as you add demonstrations, adjust retention, and switch patterns is another. The guided journey consistently helps learners grasp concepts that paragraphs of text cannot convey alone.

References

- Engdahl, B., et al. (2026). BDH-CQ: Introducing In-Context Learning with Recurrent Latent Reasoning. arXiv:2608.09888.
- Bordes, F., et al. (2024). An In-Depth Look at Gemini's ARC-AGI Capabilities and Contamination. arXiv:2407.00645.
- Akyürek, E., et al. (2024). The Surprising Effectiveness of Test-Time Training for Abstract Reasoning. arXiv:2411.07279.
- Chollet, F. (2019). On the Measure of Intelligence. arXiv:1911.01547.

Live demonstration: https://frontend-navy-gamma-81.vercel.app/
Source code: https://github.com/Astha167/ARC_Generalization_Lab